



MATHEMATICS

2014 HSC Course Assessment Task 2

Monday March 10, 2014

General instructions

- Working time – 55 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M2A – Mr Lin

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziariaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M2B – Mr Weiss

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-5	6	7	8	9	10	Total	%
MARKS	$\frac{5}{5}$	$\frac{9}{9}$	$\frac{8}{8}$	$\frac{9}{9}$	$\frac{10}{10}$	$\frac{8}{8}$	$\frac{44}{44}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. What is the value of $e^{-0.2}$ correct to 3 decimal places?

1

(A) 0.818 (B) 0.819 (C) 1.221 (D) 1.222

2. Which expression is equivalent to $\int \sqrt{5x+1} \, dx$?

1

(A) $\frac{1}{10}(5x+1)^{\frac{3}{2}} + C$ (C) $\frac{3}{10}(5x+1)^{\frac{3}{2}} + C$

(B) $\frac{2}{15}(5x+1)^{\frac{3}{2}} + C$ (D) $\frac{2}{3}(5x+1)^{\frac{3}{2}} + C$

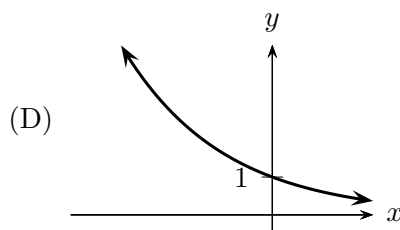
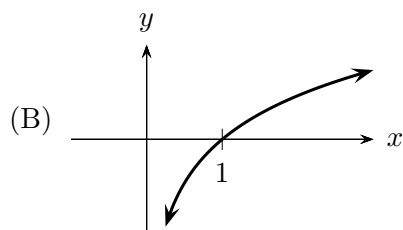
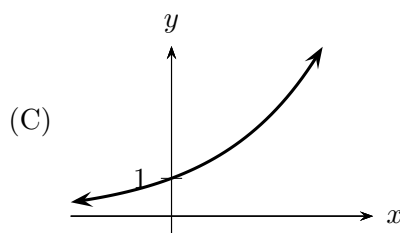
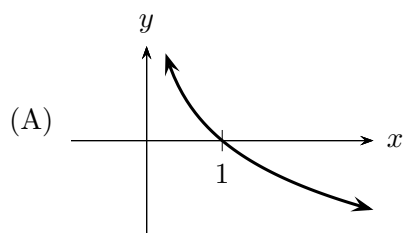
3. If $f(x) = e^{-x} - 2e^{-2x}$, what is the value of $f(\ln 2)$?

1

(A) 0 (B) 4 (C) 6 (D) 10

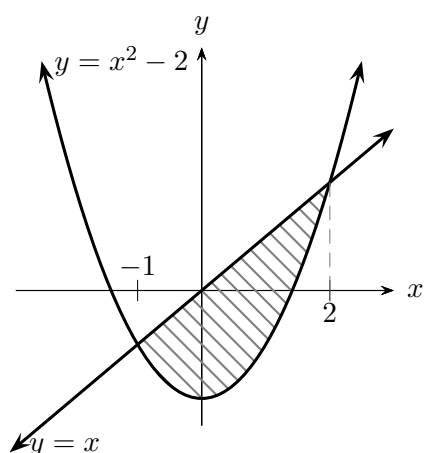
4. Which of the following best represents the shape of the graph $y = \log_e x$?

1



5. Which of the following integrals is representative of the shaded area?

1



(A) $\int_{-1}^2 (x - x^2 + 2) \, dx$

(B) $\int_{-1}^2 (x^2 + x - 2) \, dx$

(C) $\int_{-1}^2 (x^2 - x - 2) \, dx$

(D) $\int_{-1}^2 (x - x^2 - 2) \, dx$

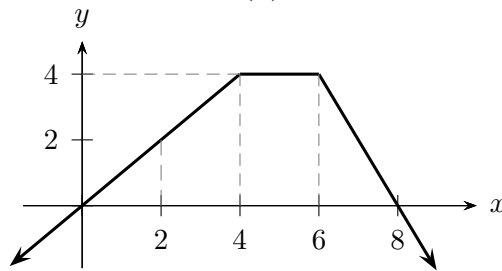
End of Section I.
Examination continues overleaf.

Question 6 (9 Marks)

Commence a NEW page.

Marks

- (a) The function
- $y = f(x)$
- is shown in the following diagram.



Evaluate:

i. $\int_0^8 f(x) dx.$

2

ii. $\int_{-2}^8 f(x) dx.$

1

- (b) Evaluate
- $\int_1^4 (3x - 2)^4 dx.$

3

- (c) Find the area enclosed by the curves
- $y = x^2 - x$
- and
- $y = 5x - x^2$
- .

3**Question 7** (8 Marks)

Commence a NEW page.

Marks

- (a) Find the primitives of the following:

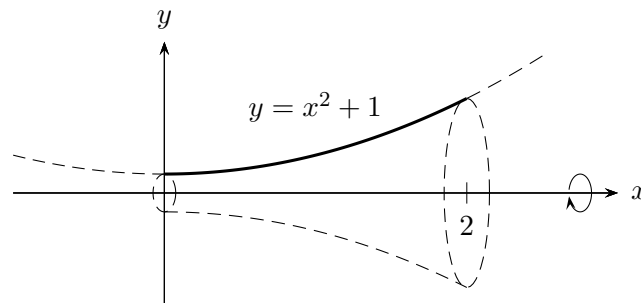
i. $\frac{x^4 - 6}{x^3}.$

2

ii. $\sqrt{x}.$

2

- (b) Find the volume of the solid of revolution generated when the curve
- $y = x^2 + 1$
- is rotated about the
- x
- axis between
- $x = 0$
- and
- $x = 2$
- .

4**Question 8** (9 Marks)

Commence a NEW page.

Marks

- (a) Given the parabola
- $x^2 = -6y$
- , find:

i. the coordinates of the vertex.

1

ii. the coordinates of the focus.

1

iii. the equation of the directrix.

1

- (b) Find the coordinates of the vertex and focus of the parabola

3

$$x^2 - 6x - 3y - 12 = 0$$

- (c) A parabola has its focus at
- $(1, -4)$
- and the directrix is the line
- $y = 6$
- .

3

Find the equation of the parabola.

- Question 9** (10 Marks) Commence a NEW page. **Marks**
- (a) Evaluate the following, giving your answer in simplest form.
- $\frac{d}{dx} \left(\frac{x}{e^x} \right).$ **2**
 - $\frac{d}{dx} (x^2 e^{-(x+2)}).$ **2**
 - $\frac{d}{dx} \left(\log_e \left(\frac{3x+4}{2x-1} \right) \right).$ **2**
- (b) For the function $y = \log_e(2x+1)$,
- Find the domain of the function. **1**
 - Find the range of the function. **1**
 - Sketch the function, showing all important information. **2**

- Question 10** (8 Marks) Commence a NEW page. **Marks**
- (a) Evaluate:
- $\int_0^2 3e^{1-5x} dx.$ **2**
 - $\int xe^{x^2} dx.$ **2**
- (b) Given $y = \log_e x$,
- | | | | | | |
|------------|---|---|---|---|---|
| x | 1 | 2 | 3 | 4 | 5 |
| $\log_e x$ | | | | | |
- Copy and complete the table above, giving the values correct to 3 decimal places. **1**
 - Use Simpson's Rule with five function values to estimate **3**

$$\int_1^5 \log_e x dx$$

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

Section I

1. (D) 2. (B) 3. (D) 4. (C) 5. (A)

Section II

Question 5 (Berry)

(a)

Marking

Q1 B

2 B

3 A

4 B

5 A

Q6 Lin

Q7 Lin

Q8 Weiss

Q9 Zuber

Q10 Ziaziaris

Q6) a) i) $2 + 6 + 8 + 4$

$$= 20$$

2

ii) $20 - 2$

$$= 18$$

1

b) $\int_1^4 (3x-2)^4 dx$

$$= \left[\frac{(3x-2)^5}{15} \right]_1^4$$

1

$$= \frac{10^5}{15} - \frac{1}{15}$$

1

$$= \frac{10^5 - 1}{15}$$

1

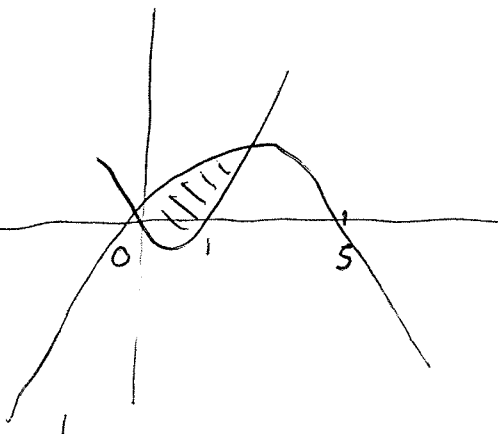
c)

$$5x - x^2 = x^2 - x$$

$$2x^2 - 6x = 0$$

$$2x(x-3) = 0$$

$$\therefore x = 0 \text{ or } 3$$



$$\int_0^3 (5x - x^2 - (x^2 - x)) dx$$

$$\int_0^3 (6x - 2x^2) dx$$

$$= \left[3x^2 - \frac{2x^3}{3} \right]_0^3$$

$$= 27 - 18 = 9 \text{ u}^2$$

$$Q7) a) i) \int x - 6x^{-3} dx$$

$$= \frac{x^2}{2} + 3x^{-2} + C$$

$$ii) \int x^{\frac{1}{2}} dx$$

$$= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$b) V = \pi \int_0^2 y^2 dx$$

$$= \pi \int_0^2 x^4 + 2x^2 + 1 dx$$

$$= \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^2$$

$$= \left(\frac{32}{5} + \frac{16}{3} + 2 \right) \pi u^3$$

$$= 13 \frac{11}{15} \pi u^3$$

Q8)

a) i) $x^2 = 4ay$

$$x^2 = -6y$$

$$a = -\frac{3}{2}$$

i) $V(0,0)$

ii) $S(0, -\frac{3}{2})$

iii) $y = \frac{3}{2}$

b) $x^2 - 6x - 3y - 12 = 0$

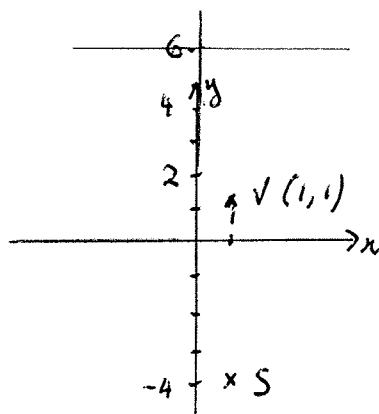
$$x^2 - 6x + 9 = \cancel{x^2} 3y + 12 + 9$$

$$(x-3)^2 = 3(y+7)$$

$\therefore$ Vertex $(3, -7)$

Focus $(3, -6\frac{1}{4})$

c)



$$(x-1)^2 = -20(y-1)$$

Q9)

a)

i/

$\frac{d}{dx}$

$\frac{x}{e^x}$

$$= \frac{e^x \cdot 1 - x e^x}{e^{2x}}$$

$$= \frac{x^x (1-x)}{e^{2x}}$$

$$= \frac{1-x}{e^x}$$

ii/

$\frac{d}{dx}$

$x^2 e^{-(x+2)}$

$$= 2x e^{-(x+2)} + x^2 e^{-(x+2)}$$

$$= x e^{-(x+2)} (2-x)$$

$$\text{iii/ } \frac{d}{dx} \left(\ln(3x+4) - \ln(2x-1) \right)$$

$$= \frac{3}{3x+4} - \frac{2}{2x-1}$$

$$= \frac{6x-3-6x-8}{(3x+4)(2x-1)}$$

$$= \frac{-11}{(3x+4)(2x-1)}$$

Q10)

a) i/ $\int_0^2 3e^{1-5x} dx$

$$= \frac{-3}{5} e^{1-5x} \Big|_0^2$$

$$= \frac{-3}{5} e^9 + \frac{3}{5} e^1$$

ii/ $\int x e^{x^2} dx$

$$= \frac{1}{2} \int 2x e^{x^2} dx$$

$$= \frac{1}{2} e^{x^2} + C$$

b) i/

x	1	2	3	4	5
$\ln x$	0	.693	1.099	1.386	1.609

ii/ $\int_1^5 \log_e x dx = \frac{1}{3} \left[0 + 1.609 + 4(.693 + 1.386) + 2(1.099) \right]$

$$= 4.041$$